CM5 socket, M.2 connector

File: cm5_socket.kicad_sch

System MCU, Watchdog, EEPROM

File: system_mcu.kicad_sch

AC power, DC power, AC sync

File: ac_dc_power.kicad_sch

Isolated digital IO, Fan controller, Encoder

File: digital_io_fan_encoder.kicad_sch

7-port USB3 Hub

File: usb3_hub.kicad_sch

Rear USB, USB-C

File: usb_conn.kicad_sch

Ethernet, Video outputs, Ref clk

File: video_outputs.kicad_sch

DIB 2.0

File: dib.kicad_sch

Trigger switch matrix, Power modules coupling

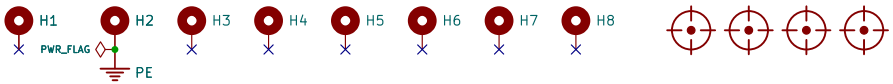
File: power_coupling.kicad_sch

USB Port	USB type	STM32H7 CM5	Raspberry PI CM5
Rear top	USB A	FS only	USB #0, HS/SS
Rear bottom	USB A	FS/HS	USB Hub P1 Flex
Front	USB C	USB Hub P2, HS/SS, Dual role	
DIB #1	DIB 2.0	USB Hub P7, HS/SS	
DIB #2	DIB 2.0	USB Hub P6, HS/SS	
DIB #3	DIB 2.0	USB Hub P5, HS/SS	
DIB #4	DIB 2.0	USB Hub P4, HS/SS	
DIB #5	DIB 2.0	USB Hub P3, HS/SS	

I2C address	Device
0x53	EEPROM M24C32
0x2F	Fan controller EMC2301
0x2D	USB3 Hub
0x3A	PTN5150A USB CC controller
0x70	ADG2188Y 8 x 8 switch array
0x43	FXL6408 8-bit I/O Expander



OSHW
HR000123

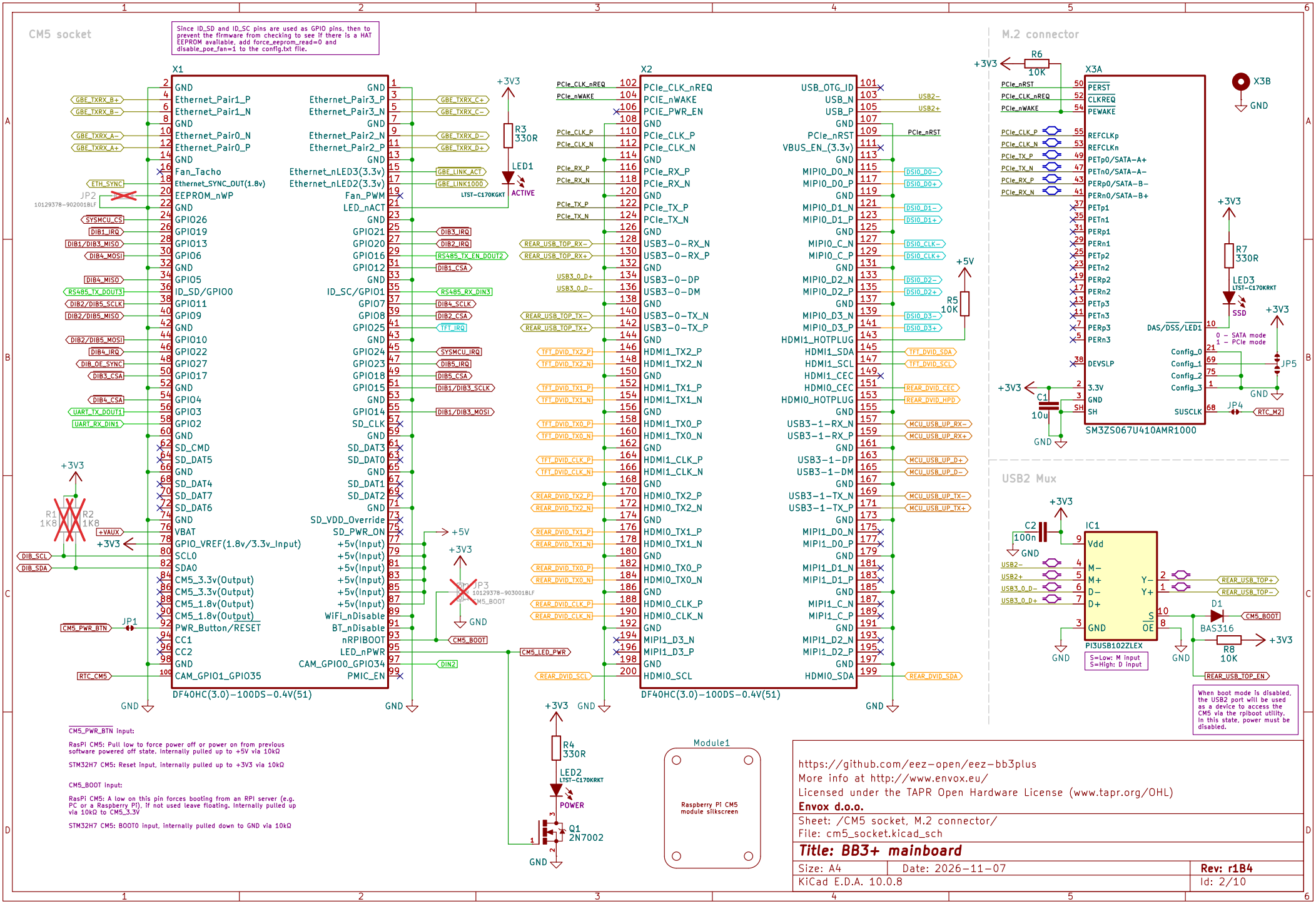


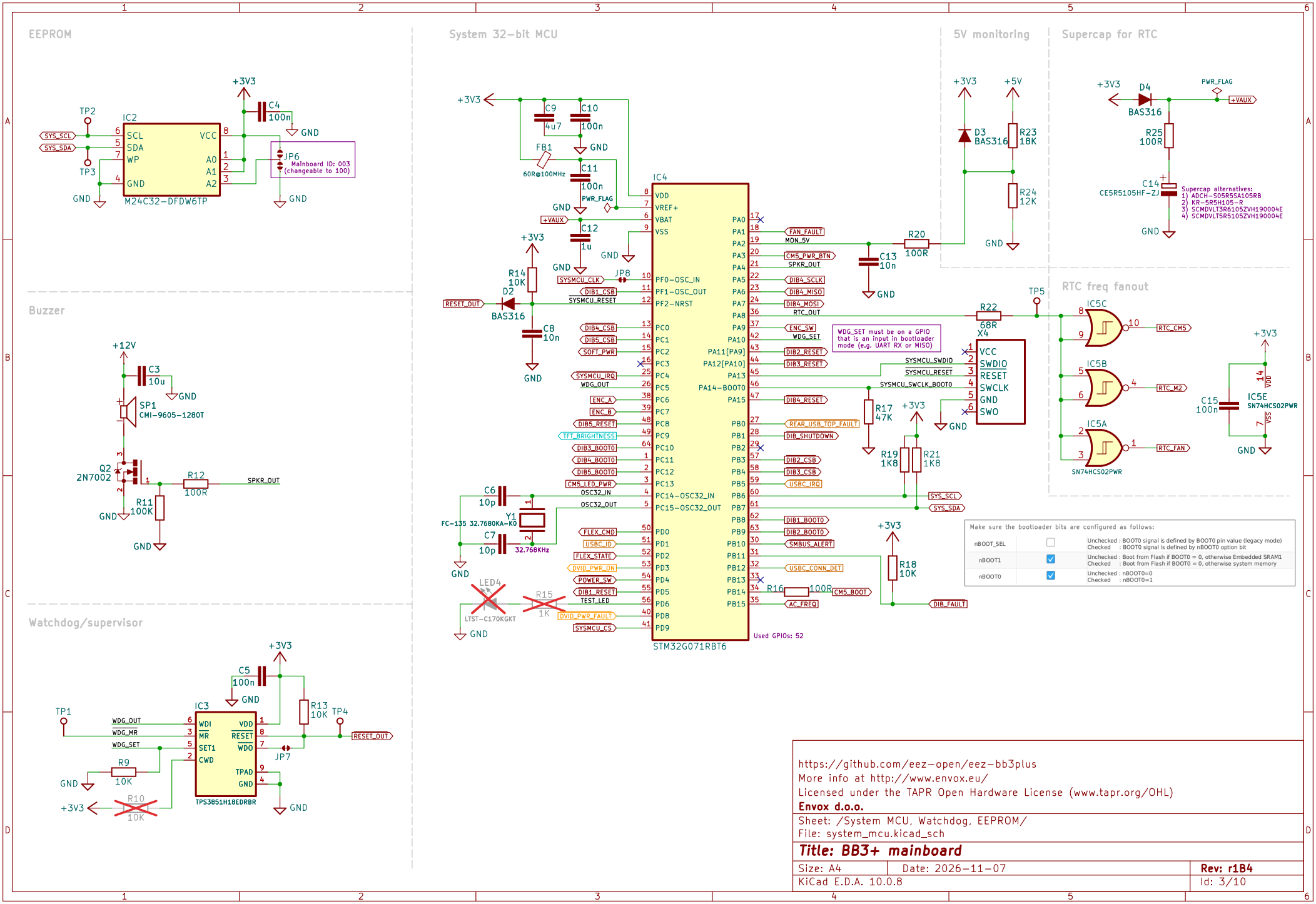
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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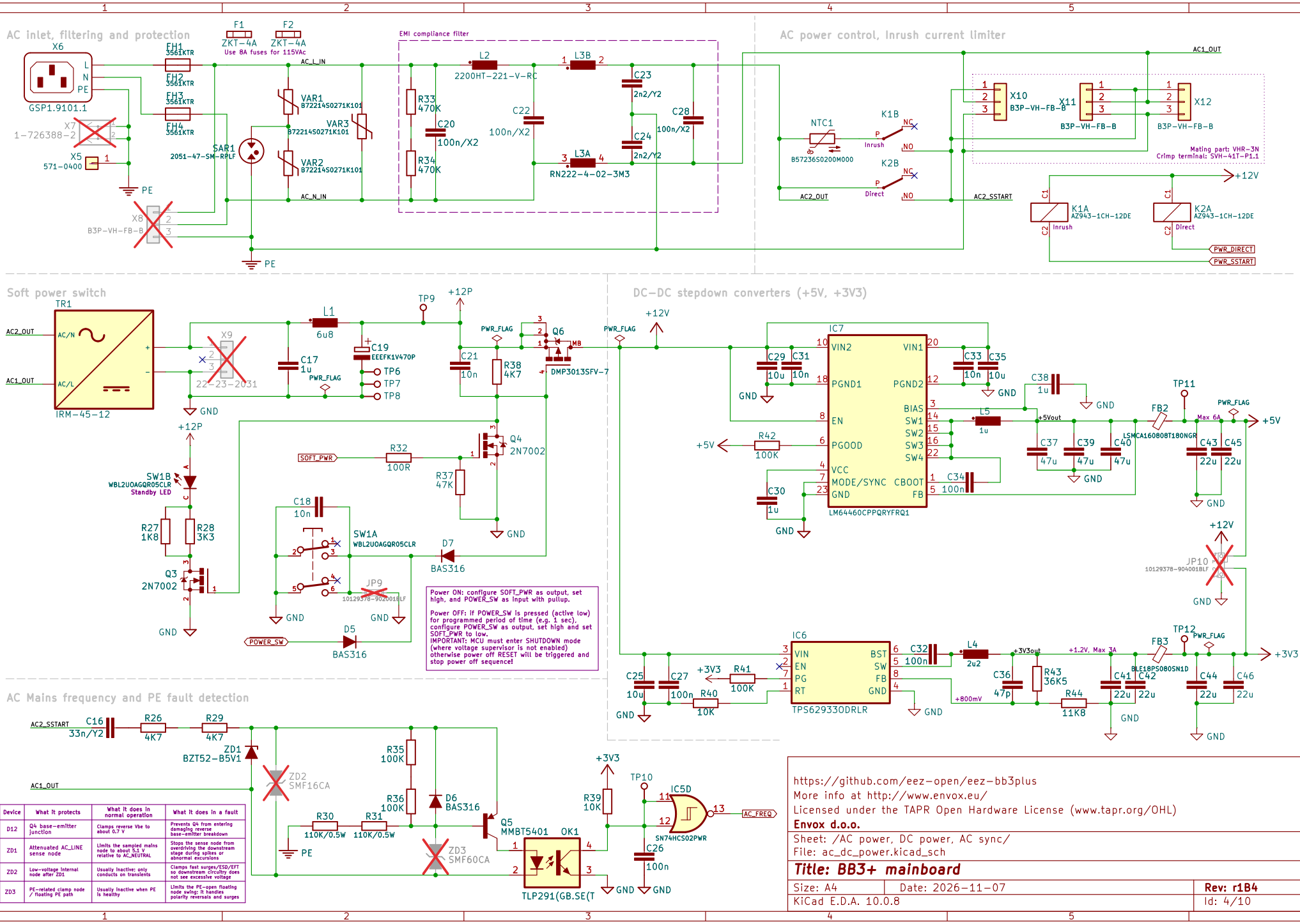
Sheet: /
File: BB3plus mainboard.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-11-07	**Rev: r1B4**
KiCad E.D.A. 10.0.8		Id: 1/10

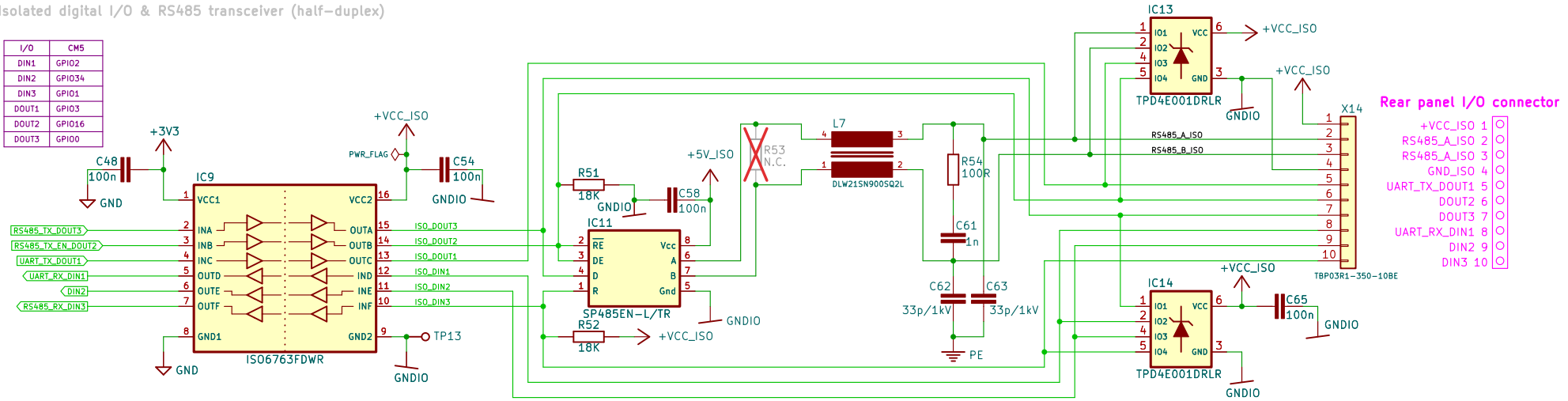

CM5 socket
Since ID_SD and ID_SC pins are used as GPIO pins, then to prevent the firmware from checking to see if there is a HAT EEPROM available, add force_eeprom_read=0 and disable_poe_fan=1 to the config.txt file.
X1
DF40HC(3.0)-100DS-0.4V(51)
X2
DF40HC(3.0)-100DS-0.4V(51)
M.2 connector
X3A
SM3ZS067U410AMR1000
X3B
USB2 Mux
IC1
PI3USB102ZLEX
S=Low: M input
S=High: D input
When boot mode is disabled, the USB2 port will be used as a device to access the CM5 via the rpiboot utility. In this state, power must be disabled.
CM5_PWR_BTN input:
RasPi CM5: Pull low to force power off or power on from previous software powered off state. Internally pulled up to +5V via 10kΩ
STM32H7 CM5: Reset input, internally pulled up to +3V3 via 10kΩ
CM5_BOOT input:
RasPi CM5: A low on this pin forces booting from an RPI server (e.g. PC or a Raspberry Pi). If not used leave floating. Internally pulled up via 10kΩ to CM5_3.3V
STM32H7 CM5: BOOT0 input, internally pulled down to GND via 10kΩ
Module1
Raspberry Pi CM5 module silkscreen
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Envox d.o.o.
Sheet: /CM5 socket, M.2 connector/
File: cm5_socket.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-11-07
Rev: r1B4
KiCad E.D.A. 10.0.8
Id: 2/10


EEPROM
IC2
M24C32-DFDW6TP
JP6
Mainboard ID: 003
(changeable to 100)
Buzzer
SP1
CMI-9605-1280T
Q2
2N7002
Watchdog/supervisor
IC3
TPS3851H18EDRBR
System 32-bit MCU
IC4
STM32G071RBT6
Used GPIOs: 52
WDG_SET must be on a GPIO that is an input in bootloader mode (e.g. UART RX or MISO)
Make sure the bootloader bits are configured as follows:
nBOOT_SEL
nBOOT1
nBOOT0
Unchecked : BOOT0 signal is defined by BOOT0 pin value (legacy mode)
Checked : BOOT0 signal is defined by nBOOT0 option bit
Unchecked : Boot from Flash if BOOT0 = 0, otherwise Embedded SRAM1
Checked : Boot from Flash if BOOT0 = 0, otherwise system memory
Unchecked : nBOOT0=0
Checked : nBOOT0=1
5V monitoring
Supercap for RTC
Supercap alternatives:
1) ADCH-S05R5SA105RB
2) KR-5R5H105-R
3) SCMDVLT3R6105ZVH190004E
4) SCMDVLT5R5105ZVH190004E
RTC freq fanout
SN74HCS02PWR
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Envox d.o.o.
Sheet: /System MCU, Watchdog, EEPROM/
File: system_mcu.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-11-07
Rev: r1B4
KiCad E.D.A. 10.0.8
Id: 3/10

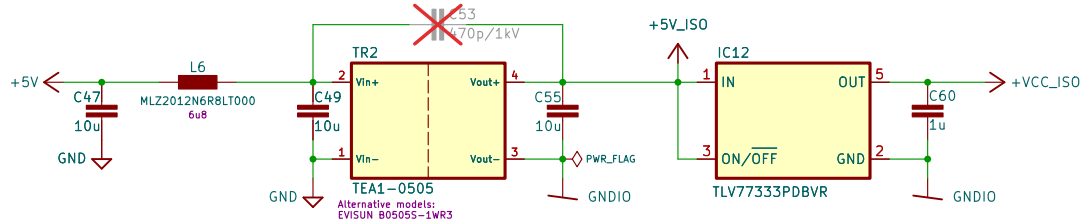

AC inlet, filtering and protection
EMI compliance filter
AC power control, Inrush current limiter
Soft power switch
DC–DC stepdown converters (+5V, +3V3)
AC Mains frequency and PE fault detection
X6
GSP1.9101.1
F1
F2
ZKT–4A
Use 8A fuses for 115VAc
FH1
FH2
FH3
FH4
3561KTR
AC_L_IN
AC_N_IN
VAR1
VAR2
VAR3
B72214S0271K101
SAR1
2051–47–SM–RPLF
L2
2200HT–221–V–RC
L3A
L3B
RN222–4–02–3M3
R33
470K
R34
470K
C20
100n/X2
C22
100n/X2
C23
2n2/Y2
C24
2n2/Y2
C28
100n/X2
X7
1–726388–2
X5
571–0400
X8
B3P–VH–FB–B
PE
NTC1
B57236S0200M000
K1B
K2B
Inrush
Direct
AC2_OUT
AC2_SSTART
AC1_OUT
X10
X11
X12
B3P–VH–FB–B
Mating part: VHR–3N
Crimp terminal: SVH–41T–P1.1
+12V
K1A
K2A
AZ943–1CH–12DE
PWR_DIRECT
PWR_SSTART
TR1
IRM–45–12
AC/N
AC/L
X9
22–23–2031
L1
6u8
C17
1u
C19
EEEFK1V470P
TP6
TP7
TP8
TP9
+12P
C21
10n
R38
4K7
PWR_FLAG
Q6
DMP3013SFV–7
GND
SW1B
WBL2UOAGQR05CLR
Standby LED
R27
1K8
R28
3K3
Q3
2N7002
SOFT_PWR
R32
100R
Q4
2N7002
R37
47K
C18
10n
SW1A
WBL2UOAGQR05CLR
JP9
10129378–902001BLF
D7
BAS316
D5
BAS316
POWER_SW
Power ON: configure SOFT_PWR as output, set high, and POWER_SW as input with pullup.
Power OFF: if POWER_SW is pressed (active low) for programmed period of time (e.g. 1 sec), configure POWER_SW as output, set high and set SOFT_PWR to low.
IMPORTANT: MCU must enter SHUTDOWN mode (where voltage supervisor is not enabled) otherwise power off RESET will be triggered and stop power off sequence!
IC7
LM64460CPPQRYFRQ1
VIN2
VIN1
PGND1
PGND2
EN
PGOOD
VCC
MODE/SYNC
GND
BIAS
SW1
SW2
SW3
SW4
CBOOT
FB
C29
10u
C31
10n
C33
10n
C35
10u
C38
1u
C30
1u
C34
100n
L5
1u
+5Vout
FB2
LSMCA160808T180NGR
Max 6A
TP11
PWR_FLAG
+5V
R42
100K
C37
47u
C39
47u
C40
47u
C43
22u
C45
22u
JP10
10129378–904001BLF
IC6
TPS629330DRLR
VIN
EN
PG
RT
BST
SW
FB
GND
C25
10u
C27
100n
R41
100K
R40
10K
C32
100n
L4
2u2
+3V3out
+1.2V, Max 3A
+800mV
FB3
BLE18PS080SN1D
TP12
+3V3
C36
47p
R43
36K5
R44
11K8
C41
22u
C42
22u
C44
22u
C46
22u
AC2_SSTART
C16
33n/Y2
R26
4K7
R29
4K7
ZD1
BZT52–B5V1
ZD2
SMF16CA
ZD3
SMF60CA
R35
100K
R36
100K
R30
110K/0.5W
R31
110K/0.5W
D6
BAS316
Q5
MMBT5401
OK1
TLP291(GB.SE(T
R39
10K
C26
100n
TP10
IC5D
SN74HCS02PWR
AC_FREQ
Device
What it protects
What it does in normal operation
What it does in a fault
D12
Q4 base–emitter junction
Clamps reverse Vbe to about 0.7 V
Prevents Q4 from entering damaging reverse base–emitter breakdown
ZD1
Attenuated AC_LINE sense node
Limits the sampled mains node to about 5.1 V relative to AC_NEUTRAL
Stops the sense node from overdriving the downstream stage during spikes or abnormal excursions
ZD2
Low–voltage internal node after ZD1
Usually inactive; only conducts on transients
Clamps fast surges/ESD/EFT so downstream circuitry does not see excessive voltage
ZD3
PE–related clamp node / floating PE path
Usually inactive when PE is healthy
Limits the PE–open floating node swing; it handles polarity reversals and surges
https://github.com/eez–open/eez–bb3plus
More info at http://www.envox.eu/
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Envox d.o.o.
Sheet: /AC power, DC power, AC sync/
File: ac_dc_power.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026–11–07
Rev: r1B4
KiCad E.D.A. 10.0.8
Id: 4/10

Isolated digital I/O & RS485 transceiver (half-duplex)

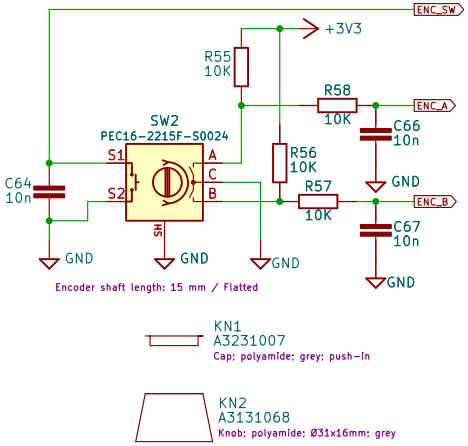
I/O	CM5
DIN1	GPIO2
DIN2	GPIO34
DIN3	GPIO1
DOUT1	GPIO3
DOUT2	GPIO16
DOUT3	GPIO0



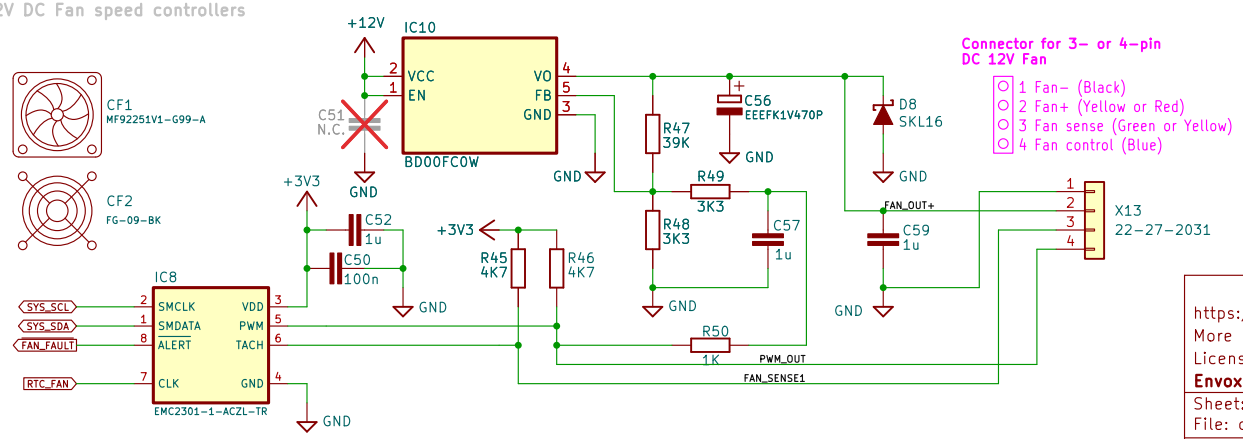
Isolated regulated power supply (+5V)



Encoder with switch



12V DC Fan speed controllers



https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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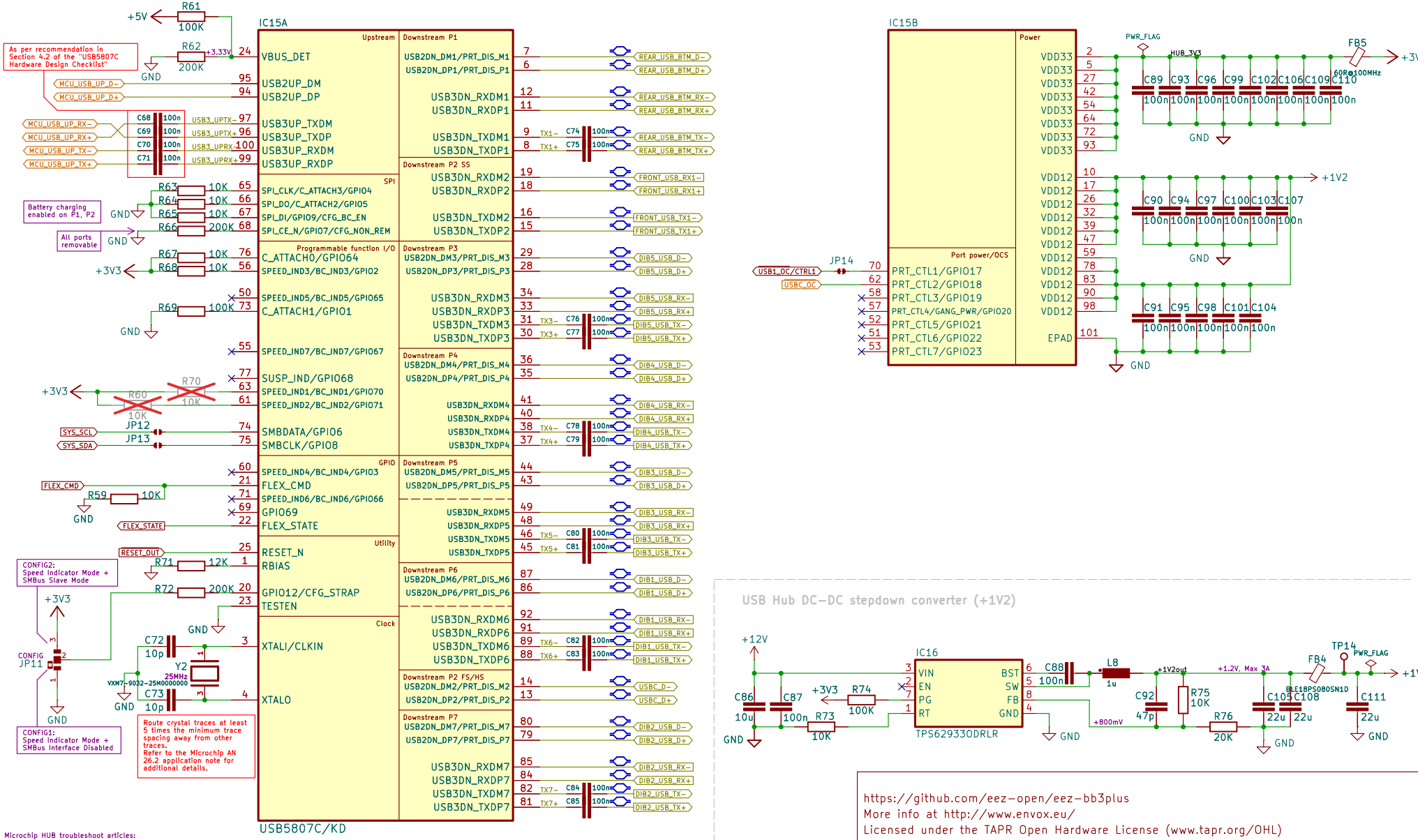
Sheet: /Isolated digital IO, Fan controller, Encoder/
File: digital_io_fan_encoder.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-11-07	**Rev: r1B4**
KiCad E.D.A. 10.0.8		Id: 5/10

7-port USB hub

All downstream ports that are not used should be disabled (use I2C to configure).



Microchip HUB troubleshoot articles:

https://support.microchip.com/s/article/USB3-Hub-Upstream-Port-Connect-Enumeration-Failure
https://support.microchip.com/s/article/USB-High-Speed-Chirp-Explained
https://support.microchip.com/s/article/USB3-Connection-Process-and-Attach-Detection

IMPORTANT:

If I2C/SMBus pull-up resistors are detected by the USB5807C at start-up, the hub will wait indefinitely to be configured by the attached I2C/SMBus master. For early prototyping, it may be necessary to physically remove the pull-up resistors until the I2C/SMBus master is fully operational and able to properly configure the hub at start-up.

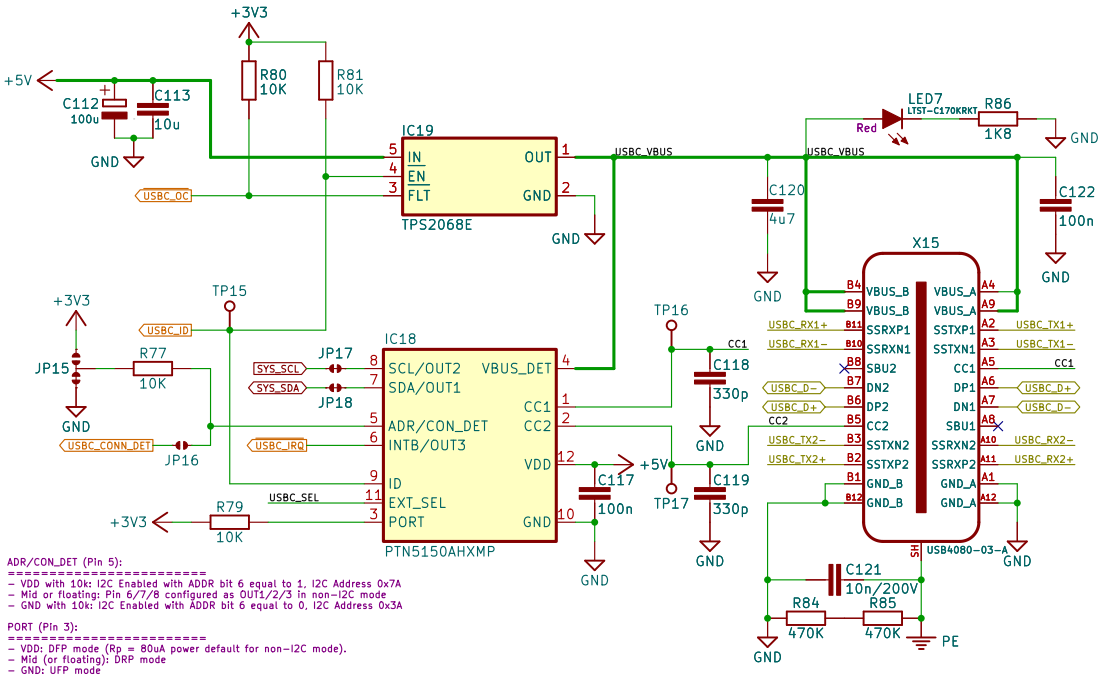
USB Hub DC-DC stepdown converter (+1V2)

https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Sheet: /7-port USB3 Hub/		
File: usb3_hub.kicad_sch		
Title: BB3+ mainboard		
Size: A4	Date: 2026-11-07	**Rev: r1B4**
KiCad E.D.A. 10.0.8		Id: 6/10

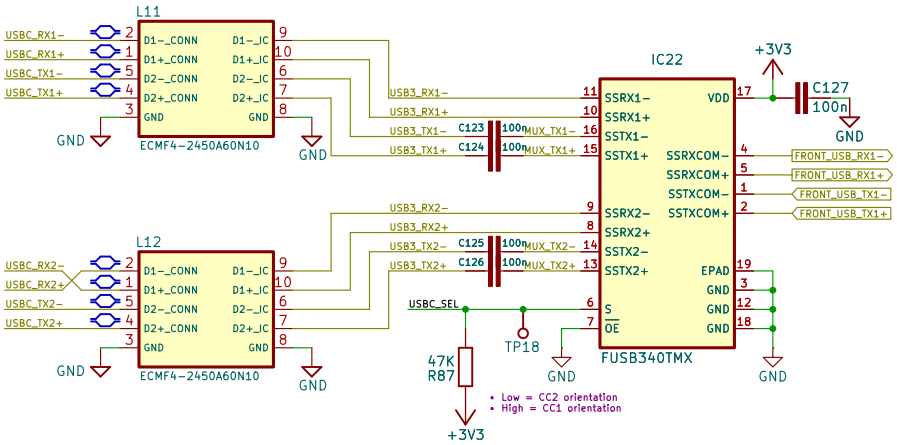
USB-C 3.0 front port (source max 1.5A)



ADR/CON_DET (Pin 5):
- VDD with 10k: I2C Enabled with ADDR bit 6 equal to 1, I2C Address 0x7A
- Mid or floating: Pin 6/7/8 configured as OUT1/2/3 in non-I2C mode
- GND with 10k: I2C Enabled with ADDR bit 6 equal to 0, I2C Address 0x3A

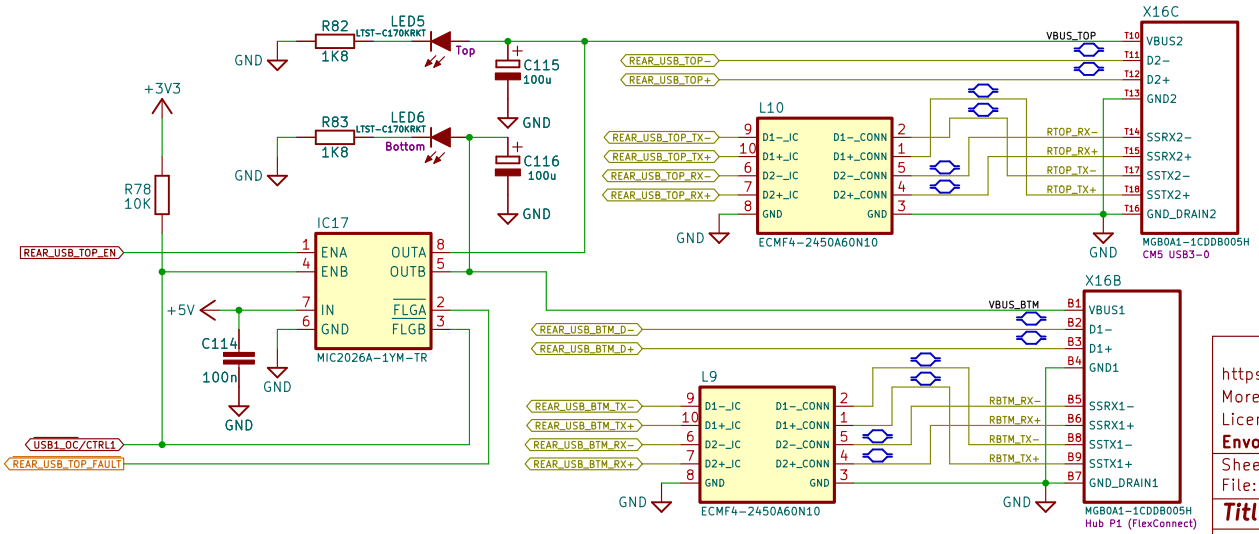
PORT (Pin 3):
- VDD: DFP mode (Rp = 80uA power default for non-I2C mode).
- Mid (or floating): DRP mode
- GND: UFP mode

USB-C 3.0 Protection and Mux

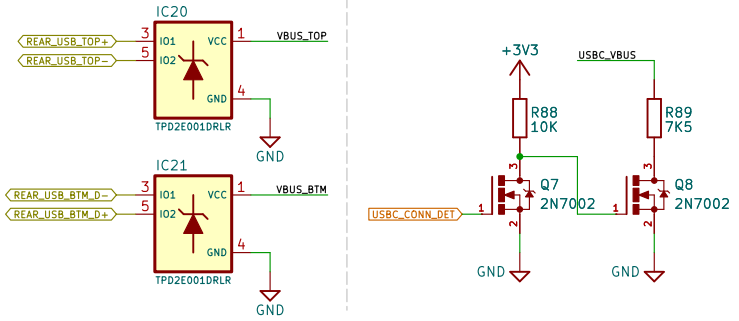


- Low = CC2 orientation
- High = CC1 orientation

USB2.0 Type-A (Rear panel)



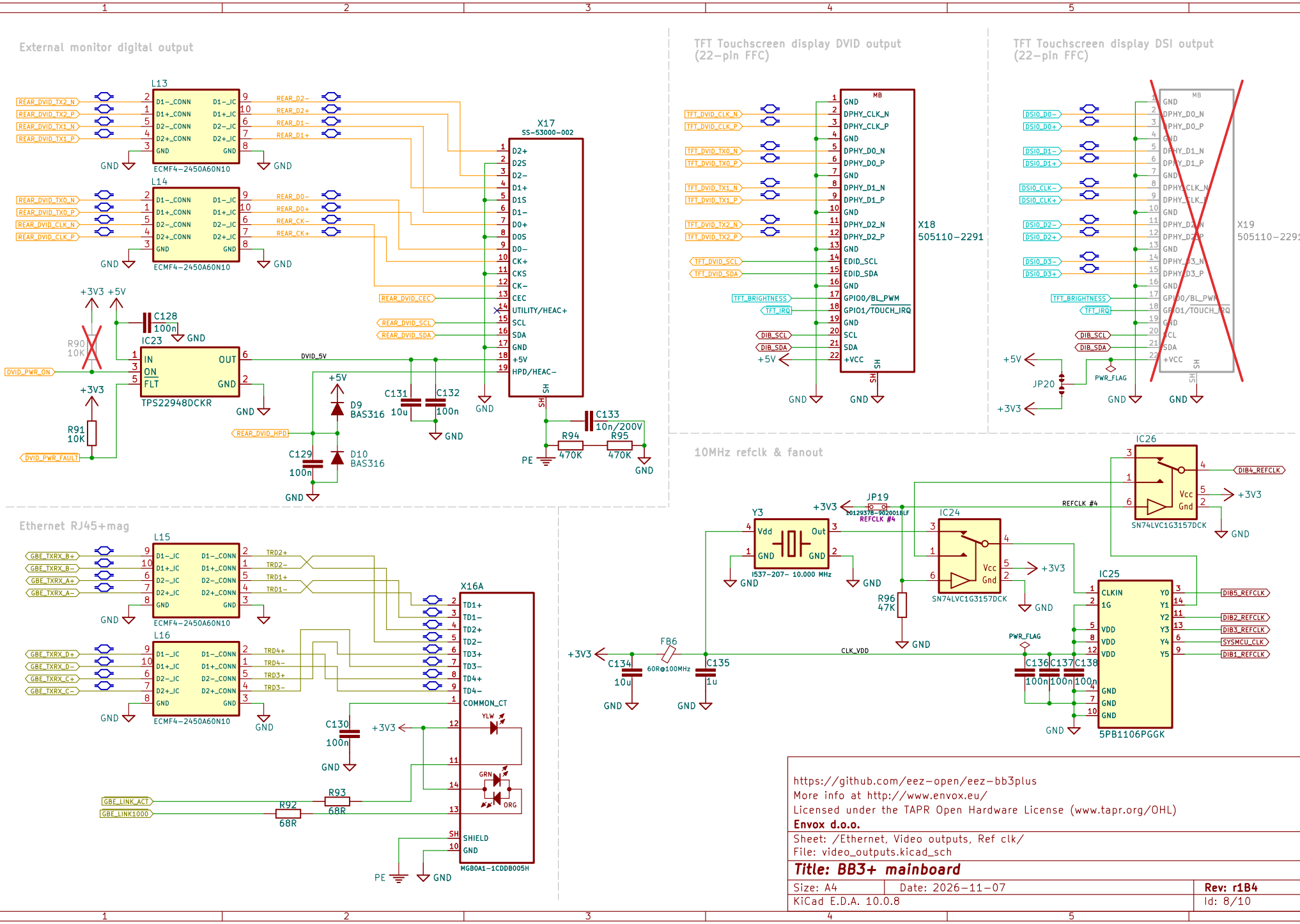
VBus Discharge Circuit

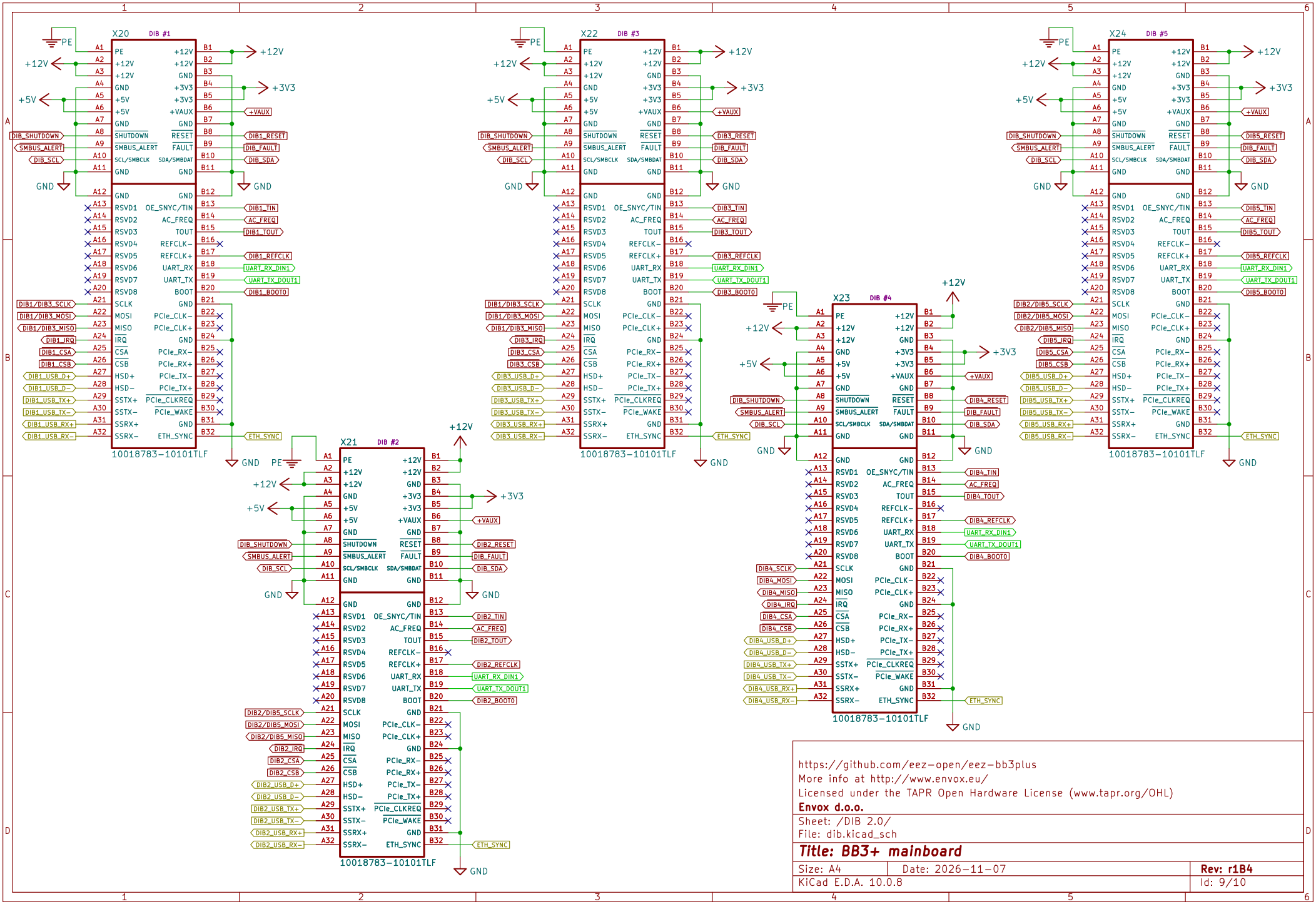


https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Envox d.o.o.

Sheet: /Rear USB, USB-C/		
File: usb_conn.kicad_sch		
Title: BB3+ mainboard		
Size: A4	Date: 2026-11-07	**Rev: r1B4**
KiCad E.D.A. 10.0.8		Id: 7/10


External monitor digital output
L13
ECMF4-2450A60N10
L14
ECMF4-2450A60N10
X17
SS-53000-002
IC23
TPS22948DCKR
C128
100n
R90
10K
R91
10K
C131
10u
C132
100n
C129
100n
D9
BAS316
D10
BAS316
C133
10n/200V
R94
470K
R95
470K
TFT Touchscreen display DVID output
(22-pin FFC)
X18
505110-2291
TFT Touchscreen display DSI output
(22-pin FFC)
X19
505110-2291
JP20
PWR_FLAG
Ethernet RJ45+mag
L15
ECMF4-2450A60N10
L16
ECMF4-2450A60N10
X16A
MGB0A1-1CDDB005H
C130
100n
R92
68R
R93
68R
10MHz refclk & fanout
JP19
2129378-9020018LF
REFCLK #4
Y3
I537-207- 10.000 MHz
IC24
SN74LVC1G3157DCK
IC26
SN74LVC1G3157DCK
R96
47K
FB6
60R@100MHz
C134
10u
C135
1u
IC25
5PB1106PGGK
C136
100n
C137
100n
C138
100n
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Envox d.o.o.
Sheet: /Ethernet, Video outputs, Ref clk/
File: video_outputs.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-11-07
Rev: r1B4
KiCad E.D.A. 10.0.8
Id: 8/10


X20
DIB #1
X21
DIB #2
X22
DIB #3
X23
DIB #4
X24
DIB #5
10018783-10101TLF
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Envox d.o.o.
Sheet: /DIB 2.0/
File: dib.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-11-07
Rev: r1B4
KiCad E.D.A. 10.0.8
Id: 9/10

DIB 2.0 Power modules coupling

20-pin DIB v2.0 Power connector

PWR+ IN 1	○ ○	2	PWR+ IN
PWR+ IN 3	○ ○	4	PWR+ IN
PWR+ IN 5	○ ○	6	PWR+ OUT
PWR+ OUT 7	○ ○	8	PWR+ OUT
PWR+ OUT 9	○ ○	10	PWR+ OUT
PWR− OUT 11	○ ○	12	PWR− OUT
PWR− OUT 13	○ ○	14	PWR− OUT
PWR− IN 15	○ ○	16	PWR− IN
PWR− IN 17	○ ○	18	PWR− IN
SENSE− 19	○ ○	20	SENSE+

IC27 FXL6408UMX
Pin compatible alternative: PI4IOE5V6408
IC28 ULN2003D
ZD4 MMSZ5230BT1G
C139 100n
+3V3
GND
SYS_SCL
SYS_SDA
RESET_OUT
PWR_SSTART
PWR_DIRECT

K3A K4A K5A K6A K7A — RT424012 PAR3 / PAR / SER / CGND / SRA
+12V

K6B K7C K7B K6C K4B K4C K5B K5C K3C K3B
CGND SRA PAR SER PAR3

X25 X26 X28 — PH2-20-UA
CH1_IN_POS CH1_OUT_POS CH1_OUT_NEG CH1_SENSE+ CH1_SENSE−
CH2_IN_POS CH2_OUT_POS CH2_OUT_NEG CH2_IN_NEG CH2_SENSE+ CH2_SENSE−
CH3_IN_POS CH3_OUT_POS CH3_OUT_NEG CH3_IN_NEG
SRA_SER

Trigger 8 x 8 Switch matrix

IC29 ADG2188Y
C140 10u, C142 100n, C141 10u, C143 100n
JP21 JP22 JP23 JP24 JP25
DIB1_TOUT DIB2_TOUT DIB3_TOUT DIB4_TOUT DIB5_TOUT AC_FREQ DIN2 DIB_OE_SYNC
DIB1_TIN DIB2_TIN DIB3_TIN DIB4_TIN DIB5_TIN RS485_TX_EN_DOUT2

Rear front power outputs

C144 10n/200V, C145 10n/200V
X27 TBP03R1-350-10BE
CH2_OUT_POS CH2_OUT_NEG CH2_SENSE+ CH2_SENSE− CH1_OUT_POS CH1_OUT_NEG CH1_SENSE+ CH1_SENSE− PE

Rear panel power connector
(Slots #1 and #2 only)

CH2_OUT+	1
CH2_OUT−	2
CH2_SENSE+	3
CH2_SENSE−	4
PE	5
CH1_OUT+	6
CH1_OUT−	7
CH1_SENSE+	8
CH1_SENSE−	9
PE	10

https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Envox d.o.o.

Sheet: /Trigger switch matrix, Power modules coupling/
File: power_coupling.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-11-07	**Rev: r1B4**
KiCad E.D.A. 10.0.8		Id: 10/10